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EVEN END SEMESTER EXAMINATION - 2025

Name of the Course: B. Tech (ME)

Semester: VIII

Name of the paper: Tribology

Paper Code: TME815

Time: 3 hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1.	20 marks	
(a)	Define tribology. Explain its interdisciplinary nature and importance in modern mechanical systems with relevant examples.	CO1
(b)	Discuss different types of viscosity. How do absolute and kinematic viscosities vary with temperature? Describe the method of determining viscosity index.	
(c)	Explain the laws of static friction and their relevance in engineering applications. Include examples from machine design.	
Q2.	20 marks	CO2
(a)	Define wear. Explain the basic mechanisms of wear with suitable diagrams and examples.	
(b)	Explain the influence of friction on wear behaviour in metallic and non-metallic materials.	
(c)	Compare the wear characteristics of metals and non-metals. What parameters influence their wear performance?	
Q3.	20 marks	CO3/CO4
(a)	Explain the Delamination theory and Oxidative theory of wear. What are their implications in real-world applications?	
(b)	Evaluate the wear resistance of different materials. How does microstructure influence wear behaviour?	
(c)	Explain the principle of hydrostatic lubrication. How is it different from hydrodynamic lubrication?	
Q4.	20 marks	CO4/CO5
(a)	Discuss the general requirements and important properties of bearing materials used in hydrostatic systems.	

(b)	Compare the performance of metallic and non-metallic bearing materials in hydrostatic lubrication.	
(c)	Describe the role of external pressure sources in hydrostatic lubrication systems. How does it influence film thickness and load-carrying capacity?	
Q5.	20 marks	CO6
(a)	Explain the principle of hydrodynamic lubrication. How is a pressure profile developed in a journal bearing?	
(b)	Derive Reynolds' equation for two-dimensional lubrication. State the assumptions clearly.	
(c)	Describe oil whip and oil whirl phenomena. How do these affect the stability of rotating machinery?	